# 1) Library Book Circulation System

## Problem Statement

Design a **Library Book Circulation System** for a small institution.

The system should manage different types of books and track which books are currently available, issued, or waiting to be processed. Each book must contain common details such as book ID, title, and author. The library currently handles at least two categories of books:

- Academic Book
- Magazine

The system must support:

- storing all books in a suitable collection
- preventing duplicate book entries based on logical identity
- maintaining a sorted view of books by title or ID
- handling a queue of book issue requests
- displaying library data in a structured way
- allowing traversal and selective removal of records when needed

The design should reflect proper object-oriented structure and should use collection types according to the behavior required.

# 2) Online Order Processing and Dispatch Manager

## Problem Statement

Design an **Order Processing and Dispatch Manager** for an online store.

Each order contains common details such as order ID, customer name, and total amount. The platform currently supports at least two types of orders:

- Regular Order
- Priority Order

The system should:

- store incoming orders
- manage dispatch processing using a queue-based approach
- avoid duplicate tracking of already processed order IDs
- maintain customer-to-order mapping
- allow different sorting views of orders using suitable comparison logic
- support iteration-based filtering/removal of invalid or cancelled orders

The solution must demonstrate object-oriented modeling and appropriate use of collections depending on the operation.

# 3) College Student Performance Tracker

## Problem Statement

Design a **Student Performance Tracker** for a college.

Each student has common attributes such as student ID, name, and department. The system currently manages at least two student types:

- Undergraduate Student
- Postgraduate Student

The system must:

- store student records
- maintain subject-wise marks or score details for each student
- avoid duplicate student entries
- maintain a ranking or sorted order of students
- group students department-wise
- provide multiple ways to sort students using natural and custom ordering
- support safe iteration while removing students who no longer meet specific eligibility conditions

Use object-oriented principles and choose collections that match the required behaviors.

# 4) Hospital Patient Token and Record System

## Problem Statement

Design a **Hospital Patient Token and Record System**.

The hospital wants to manage patient records and service flow. Each patient has common details such as patient ID, name, and age. The hospital currently handles at least two categories of patients:

- General Patient
- Emergency Patient

The system should:

- maintain all patient records
- process consultation order using token or queue logic
- ensure duplicate patient registration does not occur
- maintain doctor-wise or department-wise patient mapping
- provide a sorted representation of patients based on selected criteria
- allow safe traversal and controlled removal of discharged patients

The class structure should be object-oriented and the data structures must be chosen intentionally based on access pattern and behavior.

# 5) Transport Route and Passenger Management System

## Problem Statement

Design a **Transport Route and Passenger Management System** for a city shuttle service.

Each passenger has basic details such as passenger ID, name, and route number. The transport service currently handles at least two categories of passengers:

- Daily Passenger
- Tourist Passenger

The system should:

- register passengers
- store route-wise passenger information
- avoid duplicate passenger registration
- maintain waiting passengers for boarding using an appropriate queue structure
- support sorted views of passengers
- support custom sorting on different fields
- allow removal of specific passengers safely while traversing records

The design should show proper use of collections and object-oriented thinking.

# 6) Product Inventory and Return Handling System

## Problem Statement

Design a **Product Inventory and Return Handling System** for a retail company.

Each product has common attributes such as product ID, name, category, and price. The inventory includes at least two product types:

- Electronic Product
- Grocery Product

The system should:

- maintain product inventory
- prevent duplicate products from being stored
- maintain category-wise or supplier-wise product mapping
- provide sorted output of products
- handle returned product requests using a queue-based process
- support different sorting criteria using natural ordering and external comparison strategies
- allow safe removal of expired, invalid, or discontinued products using proper traversal mechanisms

The program should reflect clean object-oriented structure and thoughtful collection design.

# 7) Course Enrollment and Waiting List System

## Problem Statement

Design a **Course Enrollment and Waiting List System** for a training institute.

Each course participant has details such as participant ID, name, and enrolled track. The institute currently deals with at least two participant categories:

- Regular Participant
- Corporate Participant

The system should:

- manage all participant records
- prevent duplicate enrollment
- maintain batch-wise participant grouping
- manage waiting-list participants using a queue
- provide sorted views of participants based on different criteria
- demonstrate both natural and custom ordering
- allow safe iteration-based removal of participants who withdraw or become ineligible

The design must use object-oriented principles and collection types according to problem needs rather than by habit.